

Tetsuto Nagashima

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Education

University of Southern California

M.S. Computer Science, GPA 4.00

M.S. Mechanical Engineering; B.S. Physics/Computer Science; B.A. Philosophy and Physics

Relevant coursework: Trustworthy ML via Optimization, Probabilistic and Generative Models, Online Learning and Optimization

Los Angeles, CA

Expected 2027

2019-2024

Research and Experience

USC DILL Lab

Student Researcher, advised by Swabha Swayamdipta

Los Angeles, CA

Dec 2025-Present

- Researching fine-grained evaluations for LLM reasoning, especially benchmark items that entangle multiple capabilities rather than isolating one skill.
- Designing alternatives to tree-shaped capability taxonomies, including multi-capability annotations, axis-specific forests, typed graphs, and graded DAGs.
- Comparing evaluation structures with weakness-identification metrics and exploring model-native signals such as embeddings or sparse features to reduce noisy labels.

Tetsuwan Scientific

Visiting Member of the Technical Staff

San Francisco, CA

Mar 2025-May 2025

- Prototyped LLM- and vision-based systems for lab automation, including agents for translating scientific intent and labware designs into executable or structured outputs.
- Built a multimodal OCR/LLM pipeline that converted labware blueprints into structured JSON while preserving spatial relationships between features.
- Created standardized computer-vision datasets for liquid-class calibration and trained a YOLO labware classifier, diagnosing overfitting to deck backgrounds and lighting through augmentation.

USC Workshop on AI for Discovery in the Sciences

Undergraduate Research Assistant

Los Angeles, CA

Jan 2024-May 2024

- Co-developed and taught hands-on ML tutorial notebooks for PhD students and industry professionals while still an undergraduate.
- Built Colab modules on generative diffusion models, including a custom `UNet2DModel`, noise schedulers, and classifiers for detecting generated samples.
- Created a nonlinear phase-transition detection module using Learning by Confusion, including a custom sunrise dataset from GoPro time-lapse data.

University of Hawaii Outer Galaxy Group

Undergraduate Research Fellow

Honolulu, HI

May 2021-Jul 2021

- Modeled Antlia 2's orbit using Gaia EDR3 data, developing a stellar selection criterion from isochrones and proper-motion measurements.
- Applied MCMC sampling and numerical orbit integration to test whether Antlia 2 could explain observed perturbations in the Milky Way outer disc.

Selected Projects

PoisonPerf - github.com/tetsnaga/poison-perf

Data poisoning attacks in performative prediction

Spring 2026

- Implemented poisoning attacks against learners whose deployed models change future training distributions, comparing black-box, white-box, and oracle adversaries.
- Structured the code around environments, attack threat models, and repeated performative retraining algorithms to support clean experimental comparisons.

Skills

ML/AI: PyTorch, LLM evaluations, interpretability, diffusion models, computer vision, YOLO, data poisoning, empirical ML

Programming and Tools: Python, Ollama, SLURM, C++

Scientific Computing: MCMC, numerical integration, experimental design, statistical analysis